Light

GCSE Computer Science — Data Representation: Media (Set 3)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Why do computers use ASCII to store text?

- ○ **A.** To assign a unique binary code to each character
- ○ **B.** To compress the text data
- ○ **C.** To encrypt the text for security
- ○ **D.** To display text in different fonts

Q2 (1 mark)

Standard 7-bit ASCII can represent characters with codes from 0 to:

- ○ **A.** 63

- B. 99
- C. 255
- D. 127

Q3 (1 mark)

In a bitmap image, what information is stored for each pixel?

- A. The colour of the pixel
- B. The shape of the pixel
- C. The position in the file
- D. The size of the pixel

Q4 (1 mark)

An image uses 2-bit colour depth.
How many different colours can each pixel be?

- A. 4
- B. 2
- C. 8
- D. 16

Q5 (1 mark)

As well as pixel data, bitmap files typically store metadata.

Which of the following is an example of image metadata?

- ○ **A.** The image dimensions (width and height)
- ○ **B.** The colour of the first pixel
- ○ **C.** The binary value of each pixel
- ○ **D.** The actual colour data

Q6 (1 mark)

The unit Hertz (Hz) is used when measuring:

- ○ **A.** File size
- ○ **B.** Colour depth
- ○ **C.** Bit depth
- ○ **D.** Sample rate

Q7 (1 mark)

Telephone quality audio typically uses a sample rate of 8,000 Hz.
CD quality audio uses 44,100 Hz.
Which has better sound quality and why?

- ○ **A.** Telephone, because lower sample rate means less distortion
- ○ **B.** CD, because more samples capture the sound more accurately
- ○ **C.** They are the same quality

- ○ **D.** Telephone, because it uses less data

Q8 (1 mark)

An audio file uses 8-bit depth.
How many different amplitude levels can be represented?

- ○ **A.** 8
- ○ **B.** 64
- ○ **C.** 512
- ○ **D.** 256

Q9 (1 mark)

Two identical images have the same resolution. Image A uses 8-bit colour and Image B uses 24-bit colour.
Which statement is correct?

- ○ **A.** Image A has a larger file size
- ○ **B.** Image B has a larger file size
- ○ **C.** Both images have the same file size
- ○ **D.** File size cannot be compared

Q10 (1 mark)

In ASCII, the digit characters '0' to '9' have codes 48 to 57.
What is the ASCII code for the character '5'?

- A. 5
- B. 50
- C. 55
- D. 53

Q11 (1 mark)

Reducing both the sample rate and bit depth of an audio file will:

- A. Improve audio quality
- B. Reduce audio quality
- C. Have no effect on quality
- D. Only affect the file format

Q12 (1 mark)

Increasing the colour depth of a bitmap image will:

- A. Decrease the number of pixels
- B. Reduce the file size
- C. Decrease the resolution
- D. Allow more colours to be displayed

Q13 (2 marks)

In ASCII, the character 'A' has the denary code 65.
What is the ASCII code for 'L'?

- ○ **A.** 74
- ○ **B.** 76
- ○ **C.** 78
- ○ **D.** 80

Q14 (2 marks)

In ASCII, uppercase 'A' is 65 and lowercase 'a' is 97.
What is the ASCII code for lowercase 'n'?

- ○ **A.** 78
- ○ **B.** 108
- ○ **C.** 142
- ○ **D.** 110

Q15 (2 marks)

An image uses 12-bit colour depth.
How many different colours can be displayed?

- ○ **A.** 144
- ○ **B.** 1,024
- ○ **C.** 4,096

- ○ **D.** 8,192

Q16 (2 marks)

A greyscale image has dimensions of 1280 × 720 pixels with 8-bit colour depth.
What is the file size in bits?

- ○ **A.** 921,600 bits
- ○ **B.** 7,372,800 bits
- ○ **C.** 14,745,600 bits
- ○ **D.** 22,118,400 bits

Q17 (2 marks)

An audio file is recorded at 96,000 Hz sample rate with 24-bit depth for 5 seconds.
What is the file size in bits?

- ○ **A.** 2,400,000 bits
- ○ **B.** 4,800,000 bits
- ○ **C.** 11,520,000 bits
- ○ **D.** 14,400,000 bits

Q18 (2 marks)

A sound is sampled at 16,000 Hz.
What is the sample interval (time between samples)?

- ○ **A.** 0.0000625 seconds
- ○ **B.** 0.000625 seconds
- ○ **C.** 0.00625 seconds
- ○ **D.** 0.0625 seconds

Q19 (2 marks)

An image has dimensions 1920 × 1200 pixels and uses 16-bit colour depth.
What is the file size in bytes?

- ○ **A.** 2,304,000 bytes
- ○ **B.** 4,608,000 bytes
- ○ **C.** 9,216,000 bytes
- ○ **D.** 36,864,000 bytes

Q20 (2 marks)

A photograph is taken at 640 × 480 resolution and printed at poster size (60cm × 45cm).
What limitation will be visible?

- ○ **A.** The colours will appear faded
- ○ **B.** Individual pixels will be visible as squares
- ○ **C.** The file will be too large to print
- ○ **D.** The image will appear too dark

Q21 (2 marks)

Which audio settings will produce the largest file size for a 30-second recording?

- ○ **A.** 96,000 Hz sample rate, 24-bit depth
- ○ **B.** 44,100 Hz sample rate, 16-bit depth
- ○ **C.** 22,050 Hz sample rate, 24-bit depth
- ○ **D.** 48,000 Hz sample rate, 8-bit depth

Q22 (2 marks)

The ASCII code for 'R' is 82 in denary.
What is this in binary?

- ○ **A.** 1010000
- ○ **B.** 1010010
- ○ **C.** 1010100
- ○ **D.** 1010110

Q23 (2 marks)

A voice recording uses 8-bit depth instead of 16-bit depth.
What limitation will be noticeable?

- ○ **A.** High-pitched sounds will be distorted
- ○ **B.** The recording will be shorter
- ○ **C.** Quiet sounds may be lost or contain noise

- ○ **D.** The recording will play at the wrong speed

Q24 (3 marks)

A security camera captures images at 2560 × 1920 pixels with 24-bit colour.
How many images can fit on an 8 GiB storage drive?
(1 GiB = 1024^3 bytes)

- ○ **A.** Approximately 465 images
- ○ **B.** Approximately 582 images
- ○ **C.** Approximately 698 images
- ○ **D.** Approximately 931 images

Q25 (3 marks)

A stereo audio file is recorded at 96,000 Hz sample rate with 32-bit depth for 2 minutes.
What is the approximate file size in MiB?
(Stereo = 2 channels, 1 MiB = 1024^2 bytes)

- ○ **A.** Approximately 44 MiB
- ○ **B.** Approximately 88 MiB
- ○ **C.** Approximately 176 MiB
- ○ **D.** Approximately 352 MiB

Q26 (3 marks)

Image P is 800 × 600 with 32-bit colour. Image Q is 1600 × 1200 with 8-bit colour.
Which statement correctly compares their file sizes?

- ○ **A.** Image P is larger because it has more colours
- ○ **B.** Image Q is larger because it has more pixels
- ○ **C.** They are exactly the same size
- ○ **D.** Image P is larger because 32 > 8

Q27 (3 marks)

The word 'Byte' is stored using ASCII.
If 'B' = 66, what is the total of all ASCII codes for 'Byte'?

- ○ **A.** 400
- ○ **B.** 402
- ○ **C.** 404
- ○ **D.** 406

Q28 (3 marks)

Recording M uses 32,000 Hz sample rate and 16-bit depth. Recording N uses 64,000 Hz sample rate and 8-bit depth.
Both recordings are 3 minutes long. Which statement is correct?

- ○ **A.** Recording M is larger than Recording N
- ○ **B.** Recording N is larger than Recording M
- ○ **C.** Both recordings are exactly the same size

- ○ **D.** Recording M captures more frequencies than Recording N

Q29 (3 marks)

A light sensor uses 12-bit values, where 0 represents 0 lux and 4095 represents 10,000 lux.
What is the precision (smallest detectable change) of this system?

- ○ **A.** Approximately 1.22 lux
- ○ **B.** Approximately 2.44 lux
- ○ **C.** Approximately 4.88 lux
- ○ **D.** Approximately 9.76 lux

Q30 (3 marks)

A mobile app developer needs to reduce icon images from 256 KiB to under 32 KiB each without changing dimensions.
The icons are currently 256 × 256 pixels with 32-bit colour.
What is the maximum colour depth that meets this requirement?

- ○ **A.** 8-bit colour
- ○ **B.** 4-bit colour
- ○ **C.** 2-bit colour
- ○ **D.** 1-bit colour

Total: **55** marks • Time: **30 min**

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